Question

Compound $\mathbf{A}$ (structure shown in Figure 1) generates $\mathbf{B}$ with the same molecular formula as $\mathbf{A}$ under acidic conditions. $\mathbf{A}$ reacts with methanol to form two isomeric products, both of which yield compound $\mathbf{C}$ upon reduction with $\mathrm{LiAlH_4}$.

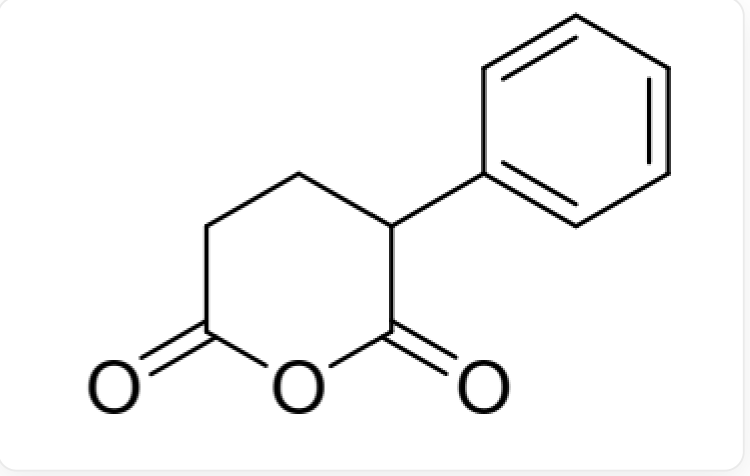

Fig. 1. The molecular structure is represented by SMILES: C1=CC=C(C=C1)C2CCC(=O)OC2=O

Several hypotheses exist regarding the structures of $\mathbf{B}$ and $\mathbf{C}$:

1. $\mathbf{B}$ contains a highly strained structure.

2. $\mathbf{B}$ can form a stable isomer under strong alkaline conditions.

3. All carbon-oxygen bonds in $\mathbf{C}$ are double bonds.

4. $\mathbf{C}$ contains only one carbon-oxygen double bond.

5. $\mathbf{C}$ contains no carbon-oxygen double bonds.

Among the following options, which one includes all correct statements and the maximum number of correct statements?

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 5

G. 1,2

H. 1,3

I. 1,4

J. 1,5

K. 2,3

L. 2,4

M. 2,5

N. 1,2,3

O. 1,2,4

P. 1,2,5

Answer

Correct Answer: M

Detailed Explanation

Compound **A** is an intramolecular anhydride and may undergo Friedel-Crafts acylation with an aryl group under acidic conditions. Considering that the resulting compound **B** has the same molecular formula as compound **A**, the acylation reaction occurs intramolecularly rather than intermolecularly. Between the two acyl groups as potential reaction sites, one would form a four-membered ring, which is highly strained and unlikely, while the other forms a six-membered ring with a lower energy barrier. Therefore, the structure of compound **B** is shown in Figure 2. The molecule does not form a highly strained four-membered ring but instead a less strained six-membered ring, so statement 1 is incorrect. Under strong base conditions, the β-carbon of the carbonyl group can be deprotonated to form a stable enolate anion, so statement 2 is correct.

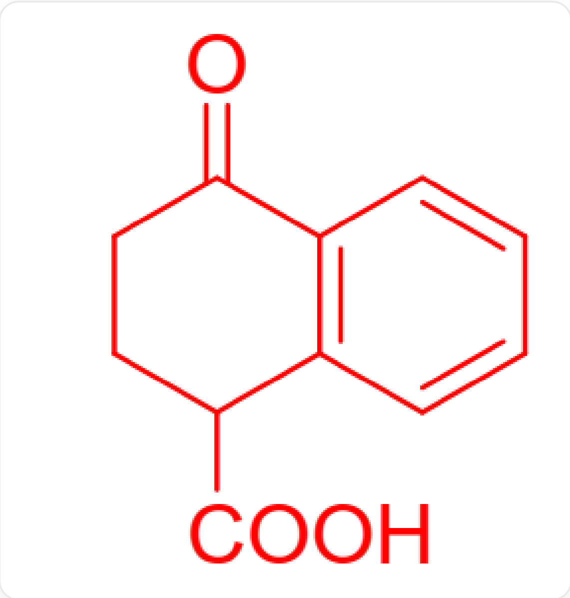

Fig. 2. The molecule in the figure is described by SMILES as: C1=CC2=C(C=C1)C(=O)CCC2C(=O)O

CHECKPOINT 1 PTS

Intramolecular Friedel-Crafts acylation forms a six-membered ring rather than a four-membered ring.

CHECKPOINT 1 PTS

The product is represented by SMILES as C1=CC2=C(C=C1)C(=O)CCC2C(=O)O.

CHECKPOINT 1 PTS

Under strong base conditions, the β-carbon of the carbonyl group can be deprotonated to form a stable enolate anion.

Compound $\mathbf{A}$ is an anhydride and undergoes alcoholysis when reacted with methanol. Since compound $\mathbf{A}$ contains two asymmetric acyl groups, alcoholysis yields two products, each containing a methyl ester group and a carboxyl group. Both groups are directly reduced to alcohols in a single step by the strong reducing agent $LiAlH_4$, resulting in the same product $\mathbf{C}$, whose structure is shown in Figure 3. The molecule contains no carbon-oxygen double bonds, only carbon-oxygen single bonds, so statements 3 and 4 are incorrect, while statement 5 is correct.

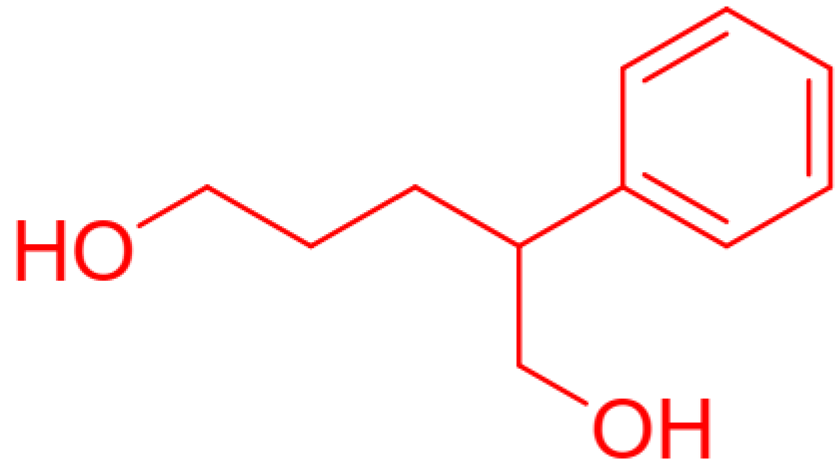

Fig. 3. The molecule in the figure is described by SMILES as: C1=CC=C(C=C1)C(CCCO)CO

CHECKPOINT 1 PTS

Methanolysis yields two products, each containing a carboxyl group and an ester group, which are directly reduced to alcohols in a single step by the strong reducing agent $LiAlH_4$, resulting in the same product.

CHECKPOINT 1 PTS

C is a diol structure, described by SMILES as: C1=CC=C(C=C1)C(CCCO)CO.

